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Comparison study of thermochemical waste-heat recuperation by steam reforming of liquid biofuels

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HIGHLIGHTS

- Thermochemical recuperation by steam reforming of liquid biofuels (methanol, ethanol, butanol, glycerol).
- An increase in process temperature leads to an increase in fuel conversion.
- The maximum of the transformation coefficient is steam reforming of ethanol and is 1.187.
- Maximum efficiency of the use of TCR is at 600 K (methanol), 700 K (ethanol), 850 K (butanol), 900 K (glycerol).

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ABSTRACT

The concept of thermochemical exhaust heat recuperation by steam reforming of biofuels is considered. Thermochemical recuperation can be considered as an on-board hydrogen production technology. A schematic diagram of a fuel-consuming equipment with thermochemical heat recuperation is described. The thermodynamic analysis of the thermochemical recuperation systems was performed to determine the efficiency of using various fuels, in particular, methanol, ethanol, n-butanol, and glycerol. The thermodynamic analysis was performed by Gibbs free energy minimization method and implemented using the Aspen Hysys program. The thermodynamic analysis was performed for a wide temperature range from 400 to 900 K, for steam-to-fuel of 1, and pressures of 1 bar. The maximum fuel conversion reaches for the following temperatures: methanol - 600 K, ethanol - 730 K, n-butanol - 860 K, glycerol - 890 K. The dependence of the reforming enthalpy on temperature is determined. It was shown that the reaction enthalpy determines the heat transformation coefficient, which shows the ratio of the low heat value of synthetic fuel and the low heat value of the initial fuel. For all studied fuels, the maximum value of the transformation coefficient is observed for steam reforming of ethanol and the maximum heat transformation coefficient is 1.187. The temperature range is determined at which the maximum efficiency of the use of thermochemical recuperation occurs due to the reforming of biofuels. For methanol, the effective temperature is about 600 K, for ethanol is about 700 K, for n-butanol is 850 K, for glycerol is more than 900 K. The results obtained make it possible to efficiently select the type of fuel for thermochemical recuperation due to steam reforming.

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Introduction

The use of various biofuels is widely used in the chemical industry [1,2], transport [3,4], energy [5,6] and other industries [7]. The simplest biofuels such as ethanol and methanol have found the greatest application in transport and industry. This is because these alcohols can be obtained from biomass, which is a renewable energy source. One of the ways to use biofuels in energy and transport is the thermochemical recuperation of the exhaust heat [5].

In recent years, many works have been published on the use of ethanol and methanol for thermochemical heat recuperation [8–10]. The main concept of thermochemical waste-heat recuperation is the transformation of exhaust heat into the chemical energy of a new synthetic fuel that has higher calorimetric properties such as low heating value [11,12]. Moreover, recently scientists and engineers from different countries have published a series of articles on thermochemical recuperation. There are papers on the experimental and theoretical study of the thermochemical waste-heat recuperations in fuel-consuming equipment: the internal combustion engines [13,14], the gas turbines [11,15], the industrial furnaces [16–18], etc. The variety of such articles shows the promise of this way of the energy efficiency increasing of fuel-consuming equipment. A comprehensive study on the use of reforming in the internal combustion engines was carried out by Tartakovsky and Sheintuch [3]. This article offers a comprehensive overview of research on fuel reforming in the internal combustion engines (ICE). The authors examined various reforming processes for methanol, ethanol, gasoline, and other fuels for the ICE application. In addition, the thermodynamics of fuel reforming in ICE and simulation approaches are also discussed. This article shows the great potential and promise of using thermochemical exhaust-heat recuperation in internal combustion engines.

The use of thermochemical waste-heat recuperation (hereinafter TCR) in the industrial plants has been widely discussed for high-temperature plants, in particular, heat-treating and glass melting furnaces. Rue and co-authors [19] showed the results of testing of a glass melting furnace with TCR in a pilot-scale process unit. The authors showed that in a furnace with TCR part of waste heat is used for the reforming of natural gas. Simulation and experimental investigation in this paper have been shown that energy recovery in the TCR systems is notably higher than is possible recuperation with either thermal regenerators or heat exchangers. For example, fuel-saving for the furnace with TCR was 25–30% higher relative to a furnace with the recuperation systems based on air preheating. This paper discussed the thermochemical recuperation by reforming of methane [20].

The use of thermochemical exhaust heat recuperation in gas turbine and combined-cycle plants has also found a wide discussion in the scientific literature. Abdallah et al. [21] investigated chemically recuperated gas turbine as a respective way to obtain ultra-low NOx emissions. Kravchenko and Verkhivker [11] calculated that the total energy efficiency of a combined cycle plant with TCR has obtained about 80–90%. Both the works mentioned above deal with TCR by steam methane reforming. However, Cao and co-authors [22] presented the results of a thermodynamic study of a gas turbine plant with TCR by dry-methane reforming. They reported that the optimized net power generation efficiency of the cycle rises to 49.6% and the exergy efficiency rises to 47.9%.

Chakravarthy et al. [10] showed the results of an analysis of ICE with thermochemical heat recuperation due to the reforming of various fuels. The authors showed that for a stoichiometric mixture of methanol and air, TCR can increase the estimated ideal engine second law efficiency by about 3% for constant pressure reforming and over 5% for constant volume reforming. For ethanol and isooctane, the estimated second law efficiency increases for constant volume reforming are 9 and 11%, respectively.

High-energy efficiency as the main advantage of TCR is highlighted for all the papers presented above. In addition, Pan et al. [23] reported that the combustion irreversibility of the new synthetic fuel is less than that of the primary fuel. In summary, thermochemical exhaust heat recuperation is the promising method to increase the energy efficiency for the various fuel-consuming equipment.

As shown above, liquid fuel has great prospects for use in the TCR systems. The use of various reforming reactions for thermochemical heat recuperation has different energy values. This is due to the fact that the performing of the reforming reaction, firstly, depends on the temperature of the process, and secondly, each reaction has a different enthalpy of reaction. For example, if the temperature of the exhaust gas is lower than 600 K, then TCR due to the ethanol reforming will have low efficiency because, at this temperature, the ethanol conversion is negligible.

Research studies on the use of TCR by liquid fuel reforming are mainly focused on the investigation of one type of fuel for TCR. Therefore, the aim of this work is the first law energy analysis of TCR through steam reforming of various liquid biofuels, in particular, methanol, ethanol, glycerol, and n-butanol. In addition, the determination of temperature ranges, in which TCR due to the reforming of various fuels will be effective, is accomplished.

Thermochemical recuperation concept

The general idea of thermochemical heat recuperation is to use the exhaust gas heat for an endothermic transformation of primary fuel. As a result of this transformation, a new synthetic fuel (product of fuel reforming) has higher low heat value (LHV) in comparison with LHV of initial fuel.

A schematic diagram of the fuel-consuming equipment with thermochemical recuperation by steam reforming of liquid fuel is shown in Fig. 1. The schematic diagram can be virtually divided into two blocks: (1) the thermochemical recuperation system (fuel reformer, steam generator (boiler), mixer (steam ejector)); (2) the fuel-consuming system (fuel-consuming equipment itself, burner assembly).

In this schematic diagram, the exhaust gas after fuel-consuming equipment is sequentially passing through the fuel reformer and the boiler (steam generator). The fuel reformer performs as a catalytic heat exchanger. The reaction mixtures before the reformer contain steam and fuel vapor.

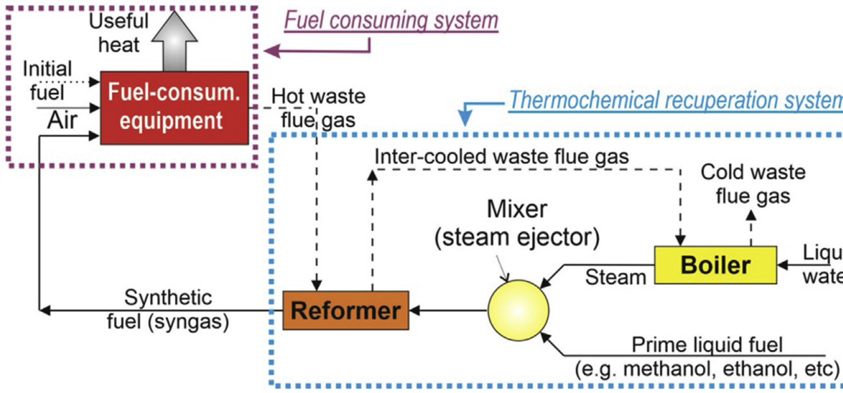

Fig. 1 – Schematic view of the fuel-consuming equipment with thermochemical recuperation by steam reforming of liquid fuel.

Inside the reaction space of the reformer [24,25], fuel and steam are reacted and to the mixture than dominantly consist of hydrogen is produced. The overall chemical endothermic reaction of the fuel reforming process can be written as follows:

$$C_nH_mO_z + yH_2O = aCO + bH_2 + \Delta H_{298}. \quad (1)$$

After the thermochemical reformer synfuel leads to the combustion chamber of the fuel-consuming equipment. In the combustion chamber the combustion process of combustion components of synfuel occurs [26–29].

The chemical reactions of the liquid fuel reforming process involve the elementary reaction of fuel reforming, the water-gas-shift reaction, and other side reactions. The mechanism of these reactions is depending on operation parameters such as pressure, temperature and steam-to-fuel ratio, as well as on the catalyst composition. The product of the fuel reforming process is synthetic fuel. This fuel after the reformer is fed to a burner for heat production in the fuel-consuming equipment.

Analyzing the schematic diagram in Fig. 1, it is clear that the heat generation from liquid fuel in the fuel-consuming equipment with thermochemical recuperation is divided into two stages. In fuel-consuming equipment with TCR, fuel is primarily converted to syngas in the reformer (first stage) and then the combustion of syngas in the burner assembly takes place (second stage). The schematic view of the heat generation from ethanol (for example) in the fuel-consuming equipment without TCR and with TCR is presented in Fig. 2.

It is known that the enthalpy of reaction (1), fuel conversion and synthesis gas composition in an equilibrium condition depends on operational parameters, such as temperature, pressure, and composition of the initial mixture. A change in the reaction enthalpy will lead to a change in the heat balance throughout the TCR system. In addition, the reaction enthalpy, the fuel conversion, and the composition of the synthesis gas substantially depend on the type of liquid fuel and technological parameters such as temperature, pressure, and inlet steam-to-fuel ratio.

The general goal of this study is the calculation of the efficiency of the thermochemical recuperation systems for various fuels. Effect of temperature and type of fuel on the reaction enthalpy and the composition of synthetic fuel in equilibrium can be established by thermodynamic analysis. The steam reforming of various liquid fuels is thermodynamically analyzed.

Methodology

The equilibrium gas composition, fuel reforming and enthalpy of fuel reforming reaction were determined via thermodynamic analysis. The algorithm of the thermodynamic analysis by the Gibbs free energy minimization (GFEM) was introduced

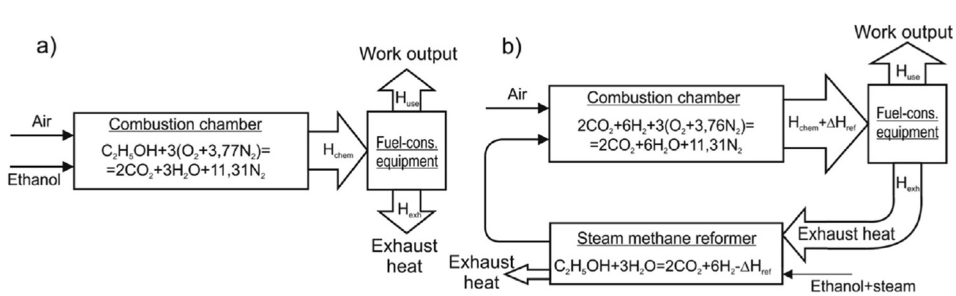

Fig. 2 – The schematic diagram of the heat generation from ethanol in the fuel-consuming equipment without thermochemical recuperation (a) and with thermochemical recuperation (b).

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in the author's previous publications [5,30,31] and it was also described in details by other authors [32–34]. Thermodynamic analysis of fuel reforming via GFEM method was introduced to determine equilibrium compositions of the product gas. The GFEM technique is based on the principles that the analyzed chemical system is thermodynamically auspicious if total Gibbs free energy has the lowest value and differential of Gibbs free energy is zero for presented operation conditions [35]:

$$dG^t = 0 \tag{2}$$

where G^t is total Gibbs free energy.

At moderate low pressure (before 5–10 bar) and high temperature, the system can be considered as ideal [36]. Moreover, the reaction products from the reformer are determined, based on the following admissions: steady flow process with the negligible change of kinetic and potential energies; perfect insulation of reformer with negligible heat loss; chemical reactions with known reactants and products compositions are equilibrium.

Total Gibbs free energy of a reforming system is presented by the sum of the chemical potential of all chemical elements:

$$G^t = \sum n_i \mu_i \tag{3}$$

where n_i is number of moles of species i; μ_i is chemical potential of species i.

The chemical potential of each element is determined as follow:

$$\mu_i = \Delta G_{fi}^0 + RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{4}$$

where G_{fi}^0 is standard Gibbs free energy of formation of species i; R is molar gas constant; T is temperature of the system; f_i is fugacity of pure species i; f_i^0 is fugacity of pure species i at the standard state.

If combine the expressions above, the equation for total Gibbs free energy of the chemical system can be written as:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{5}$$

The thermodynamic analysis based on the Gibbs free energy minimization method (GFEM) was done via special software Aspen-HYSYS [37–40]. This program is a powerful process simulator for modeling steady-state process systems where chemical reactions take place. Aspen-HYSYS is capable to simulate a single-phase or multiphase systems at equilibrium composition. A Gibbs reactor in this software is used to determine the composition of synthesis gas at the reformer outlet in reaction system under the conditions of Gibbs free energy minimization. The Gibbs reactor allows calculating the composition of synthesis gas by Gibbs free energy minimization of each component in the reaction system according to the reaction network. Fig. 3 shows the computational scheme of the thermochemical recuperation process for Aspen HYSYS.

Carbon formation and deposition is another factor that have to be discussed especially at low steam-to-fuel ratio and moderate high temperatures. However, there is sufficient information in literature that carbon formation can be controlled by improving the coking resistance of the catalysts as well as by regulating the operating conditions such as steam-to-carbon and fuel-to-oxidant ratio. In this study, oxygen-to-carbon ratio in all different feed gas compositions is either equal to or greater than unity, which indicates that there is enough oxygen to oxidize and prevent formation of solid carbon. Hence, carbon formation was omitted from equilibrium expressions.

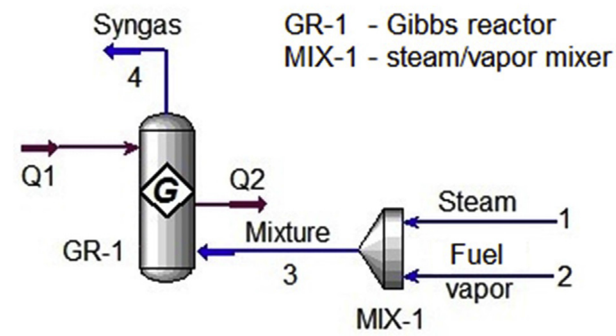

Fig. 3 – The computational scheme of the TCR process for Aspen HYSYS.

Results and discussion

Comparison study of thermochemical waste-heat recuperation by steam reforming of liquid biofuels was performed for five the most common biofuels: methanol, ethanol, n-butanol, glycerol. The effectiveness of the thermochemical recuperation systems is determined by the enthalpy of the reforming reaction. In turn, the enthalpy of the reforming reaction is determined by the initial fuel conversion. If the initial fuel conversion is zero then thermochemical heat recuperation is absent.

Fuel conversion

Comparison study of thermochemical waste-heat recuperation by steam reforming of liquid biofuels was performed for five the most common biofuels: methanol, ethanol, n-butanol, glycerol. The equilibrium fuel conversion is determined as:

$$X_{fuel} = \frac{Fuel_{in} - Fuel_{out}}{Fuel_{in}}. \tag{6}$$

where $Fuel_{in}$, $Fuel_{out}$ number of moles of fuel at the reformer inlet (in) and outlet (out).

According to the definition, the fuel conversion varies from 0 to 1. If the fuel conversion is 0, then the initial fuel is not involved in the reaction; if the conversion is 1, then all fuel has reacted with steam. For further analysis, the steam-to-fuel ratio of 1 is considered.

Fuel conversion is highly dependent on process temperature. This is explained by the fact that, as a rule, reforming reactions are reversible and at moderately low temperatures the equilibrium of the reactions is shifted to the side of

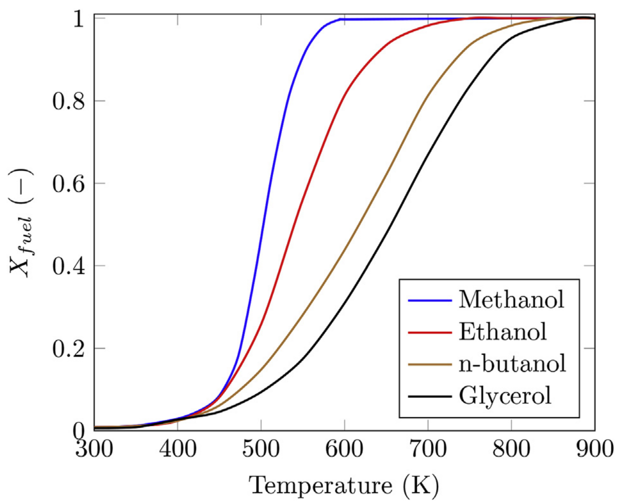

Fig. 4 – The fuel conversion as a function of temperature at p = 1 bar and fuel-to-steam ratio of 1.

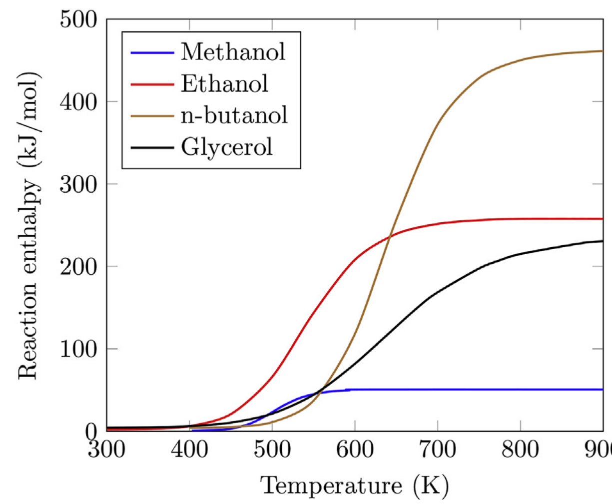

Fig. 5 – Reaction enthalpy for a steam reforming process as a function of temperature.

consuming of reforming products. Fig. 4 shows the change in the fuel conversion as a function of temperature.

Fig. 4 shows the change in fuel conversion as a function of temperature. Because the reforming reactions are endothermic, fuel conversion increases with increasing temperature. Fig. 4 shows that fuel conversion is significantly different for various fuels. In the temperature range from 500 to 700 K, there is a significant difference in fuel conversion. For glycerol at T = 600 K the conversion is 0.31 but at the same temperature, the methanol conversion is close to unity. For all the fuels in the temperature range above 900 K, the fuel conversion is close to unity.

Fig. 4 also shows the significant effect of temperature on fuel conversion, especially for methanol and ethanol. For example, in the temperature range of about 150°, the methanol conversion varies from 0.03 to 0.99. For ethanol, the temperature range in which the conversion reaches 1 from 0.1 is approximately 250°. At the same time, to increase the glycerol conversion from 0.1 to 1, a temperature range of about 400 K is required.

Fuel conversion is an important characteristic when analyzing the efficiency of thermochemical heat recuperation systems because fuel conversion shows the fundamental possibility of using thermochemical heat recuperation. If the fuel conversion is minimal, then the efficiency of thermochemical heat recuperation is minimal. However, fuel conversion is not an objective characteristic of the efficiency of using thermochemical heat recuperation. This is because at moderately low temperatures there is a thermal decomposition of methanol, ethanol, n-butanol, as well as many other adverse reactions that can be exothermic or endothermic. Therefore, to analyze the effectiveness of the TCR systems, it is necessary to know the enthalpy of the reforming process. According to the schematic diagram in Fig. 2, the enthalpy of the fuel reforming process is the exhaust heat, which is transformed into the chemical energy of new synthetic fuel. Fig. 5 shows the dependence of the enthalpy of the reforming reaction of the studied fuels at p = 1 bar and the fuel-to-steam ratio of 1.

Fig. 5 shows the change in the reaction enthalpy as a function of temperature at a pressure of 1 bar and the steam/fuel ratio = 1. From Fig. 5 it can be seen that the enthalpy of the reaction varies significantly for different fuels. The minimum enthalpy of the steam reforming reaction in the temperature range above 600 K occurs for methanol. The maximum reaction enthalpy in the temperature range above 600 K occurs for n-butanol.

The enthalpy of reaction shows the exhaust heat that has been transformed into the chemical energy of new synthetic fuel (according to the diagram in Fig. 1). Therefore, from Fig. 5 it can be concluded that 1 mol of n-butanol allows transforming the maximum amount of exhaust heat into the chemical energy of new synthetic fuel. Such an increase in the chemical energy of synthetic fuel is expressed in an increase in the low heat value-LHV (heat combustion) of the new synthetic fuel according to the energy conservation.

$$LHV_{synfuel} = LHV_{fuel} + \Delta H_{298}, \tag{7}$$

where $LHVsynfuel$, LHV_{fuel} – low heat value of synthetic fuel and initial fuel, respectively; ΔH_{298} – enthalpy of the reforming process.

The significant difference in the reaction enthalpy in Fig. 5 is because various fuels with different compositions take part in the reforming reaction. Initial fuels before reforming have different LHV. Therefore, to assess the fuel efficiency in a thermochemical recuperation system, it is necessary to know how many times the LHV of new synthetic fuel is greater than the LHV of the initial fuel. The ratio of LHV of new synthetic fuel to LHV of the original fuel is called the heat transformation coefficient. The heat transformation coefficient can be expressed as follows:

$$n_q = LHV_{synfuel}/LHV_{fuel}. \tag{8}$$

If the reforming reaction is endothermic, then the heat transformation coefficient is greater than unity. Fig. 6 shows a comparison of the heat transformation coefficients for various fuels at different temperatures.

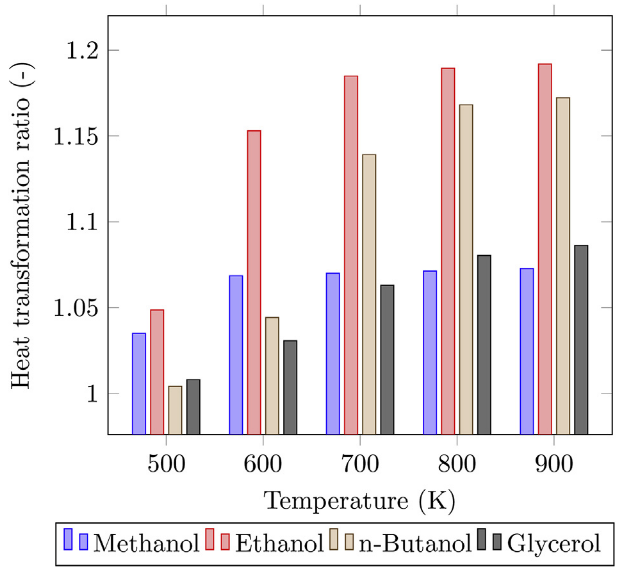

Fig. 6 – The transformation coefficients for various fuels vs. temperature.

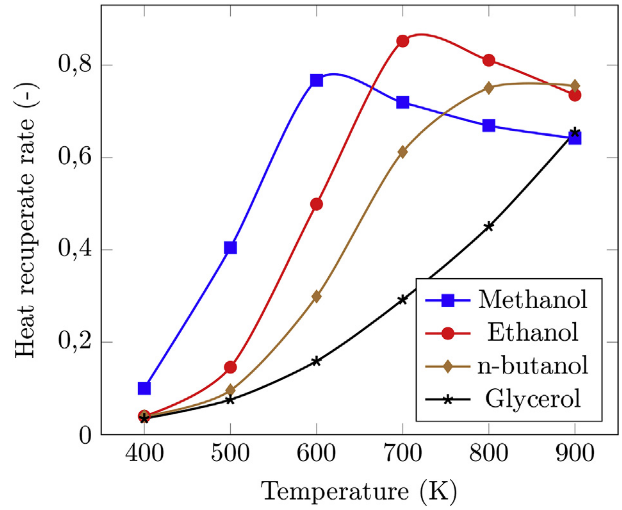

Fig. 7 – The recuperation rate that could be achieved in the reformer as a function of process temperature.

Fig. 6 makes it clear that the increasing reforming temperature increases the quantity of the heat obtained by the combustion of the synthesis gas in comparison to the combustion of initial biofuels. This is because with an increase in the temperature of the process, the reaction enthalpy increases. Fig. 6 shows that ethanol has a maximum heat transformation coefficient for the entire studied temperature range from 500 to 900 K. At the same time, at a temperature of 500 K, the value of the heat transformation coefficient for ethanol and methanol is close, and at a temperature of 900 K, the transformation coefficient for ethanol and n-butanol is close.

For ethanol, the maximum value of the transformation coefficient is 1.187. Therefore, the LHV of the synthesis gas obtained as a result of ethanol reforming is 1.187 times greater than the LHV of the initial ethanol. At the same time, even though the maximum reaction enthalpy in the temperature range above 800 K is observed for n-butanol reforming, the heat transformation coefficient remains relatively small. This is due to the fact that n-butanol has a high calorific value of 2676 kJ/mol and, in expression (8), the denominator does not allow reaching a high η_q value.

The efficiency of the application of the TCR systems by steam fuel reforming is depending on the heat transferred in the reformer for the fuel reforming reactions and syngas heating. If steam fuel reforming doesn't occur in the reformer, then the efficiency of using the TCR system equals zero or even has a negative effect because exhaust heat is used for the generation of steam. An important criterion for the effectiveness of the TCR systems is the heat recovery rate in the reformer.

The heat recovery rate in the reformer is the ratio of the heat recovered in the reformer to the heat of the exhaust flue gases. In other words, the heat recovery rate indicates how much heat from the flue gas can be recovered to the fuel cycle in the reformer. A detailed algorithm for determining the heat recovery rate in the thermochemical heat recuperation systems is described in the previous author's works [12,30].

According to the definition, the heat recovery rate that could be achieved in the reformer can be expressed as follows:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{\Delta H_{298} + \Delta H_{sg}}{H_{exh}} \tag{9}$$

where H_{rec} - a recovered heat in the TCR system; H_{exh} - an exhaust heat; ΔH_{sg} – a heat for an increasing enthalpy of synthesis gas.

Fig. 7 shows the change in the heat recovery rate that is achieved in the reformer when various fuels are used.

Fig. 7 shows that with increasing temperature, the heat recovery rate increases. At moderately low temperatures up to 500 K for ethanol, n-butanol, and glycerol, recuperation is carried out mainly by heating the synthesis gas. This is because at these temperatures the reaction enthalpy is minimal. With increasing temperature, the recovery rate increases. This is since both the reaction enthalpy increases and the heat for heating of synthesis gas increases. However, after reaching the maximum value, the recovery rate decreases. This is because the reaction enthalpy ceases to increase. In this case, the denominator in expression (9) increases. Therefore, a decrease in the recovery rate is observed. This decrease is especially evident for methanol and ethanol. It should be expected that at higher temperatures (more than 900 K) a similar character of the dependences takes place for n-butanol and glycerol.

The maximum recovery rate of 0.831 is observed for the TCR system due to the steam reforming of ethanol at a temperature of about 700 K. Fig. 7 makes it possible to determine the temperature range at which there is a maximum efficiency of using of thermochemical recuperation by steam reforming of various biofuels. According to Fig. 7, for methanol, this temperature is about 600 K, for ethanol about 700 K, for n-butanol 850 K, for glycerol more than 900 K. The results in Fig. 7 allow to efficiently select the type of fuel for thermochemical recuperation by steam reforming.

Conclusion

The thermochemical heat recuperation systems by steam reforming of biofuels are considered. The thermochemical recuperation concept is discussed. Four types of biofuels are analyzed such as methanol, ethanol, n-butanol, and glycerol. The thermodynamic analysis of the thermochemical waste-hear recuperation systems realized via various fuel reforming processes is performed to determine its effectiveness. The thermodynamic analysis is based on Gibbs free energy minimization method and accomplished via Aspen HYSYS software.

An increase in process temperature leads to an increase in fuel conversion. At the same time, fuel conversion reaches its maximum value at the following temperatures: methanol - 600 K, ethanol - 730 K, n-butanol - 860, glycerol - 890 K. The article shows that the enthalpy of the reforming process determines the effectiveness of thermochemical recuperation systems. The reforming enthalpy also substantially depends on the process temperature. Moreover, the maximum enthalpy of the process for reforming n-butanol is more than 5 times greater than for reforming methanol. In addition, the reaction enthalpy determines the heat transformation coefficient, which shows the ratio of the low heat value of synthetic fuel and the low heat value of the original fuel. For all studied fuels, the maximum value of the transformation coefficient is observed for steam reforming of ethanol and the maximum heat transformation coefficient is 1.187. Therefore, the low heat value of the synthesis gas obtained as a result of ethanol reforming is 1.187 times greater than the low heat value of the initial ethanol. It is shown that an important criterion for the effectiveness of the TCR systems is the heat recovery rate in the reformer. The maximum recovery rate of 0.831 takes place for TCR due to steam reforming of ethanol at a temperature of about 700 K. The temperature range is determined at which the maximum efficiency of the use of TCR occurs due to the reforming of biofuels. For methanol, the effective temperature is about 600 K, for ethanol about 700 K, for n-butanol 850 K, for glycerol more than 900 K. The results obtained make it possible to efficiently select the type of fuel for thermochemical recuperation due to steam reforming. The results of this paper can be used for the first law energy analysis of the thermochemical waste-heat recuperation system by steam reforming of liquid biofuels.

Acknowledgment

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